Quantum Mechanics II — Exercise Sheet 10

Bound states and the continuum

Variant: Questions only

Exercise 1 (Counting square-well levels at a threshold). For $V = -V_0 \mathbf{1}_{[-w,w]}$, put $\gamma = w\sqrt{2mV_0}/\hbar$. Use the lecture's even and odd matching equations to count strictly negative levels for $\gamma = \pi/2$, $\gamma = \pi$ and $\gamma = 5\pi/4$. Explain how a zero-energy threshold affects the count.

Exercise 2 (The deep well and its reference energy). Keep w fixed and let $V_0 \to \infty$ in the preceding well. For a fixed bound-state index $n \geq 1$, measured from the bottom, find the limit of $E_n(V_0) + V_0$. Explain why the unshifted $E_n(V_0)$ has no finite Dirichlet-box limit. Use the matching equations to justify the limiting wall condition.

Exercise 3 (A smooth trial state). For the square well take $R > 0$ and $f_R(x) = A_R/[1 + (x/R)^2]$, with $A_R > 0$ chosen to normalize it. (i) Compute A_R and $\langle f_R, Hf_R\rangle$. (ii) Show that this energy is negative for sufficiently large R for every $V_0 > 0$. (iii) Verify that $f_R \in D(H_0)$, so the lecture's variational principle applies.

Exercise 4 (An added repulsive potential). Let $V \geq 0$ be continuous and compactly supported, and $H = H_0 + V$ on $D(H_0)$. Show that H has no negative spectrum and that every $\lambda \geq 0$ lies in its spectrum. Use the lecture's shifted Weyl packets for the second claim.